Functional Specifications: Smart Analysis Framework

1. Executive Summary

The Smart Analysis Framework is an orchestration layer designed to transform raw document assets into structured, expert-level analyses. It separates the **Execution** (Workbench), the **Configuration** (Template Studio), and the **Governance** (Admin Interface). This document provides exhaustive specifications for developers and stakeholders to build a consistent, high-precision analysis engine.

2. Workbench: Analysis Execution (End-User View)

The Workbench is integrated into the existing Canvas environment. It acts as the consumption layer for the templates created in the Studio.

2.1 Trigger & Contextual Selection

- **Action:** The user selects one or more documents on the Canvas and right-clicks to open a contextual menu.
- **Template Filtering:** The system dynamically filters available templates based on the selection count.
- **Execution Modal:** If a template requires specific parameters, a non-intrusive modal appears.

2.2 Result Visualization

- **Textual:** High-fidelity Markdown (.md).
- **Tables:** Structured Markdown or CSV.
- **Diagrams:** Interactive SVG/PNG rendered from Mermaid code.
- **Traceability:** Clickable source links with document highlighting based on Evidence Objects.

3. Template Studio: The Configuration Engine (Power User View)

The Template Studio is a No-Code environment where Power Users build sequential **Analysis Pipelines**.

3.1 Pipeline Orchestration

Analyses are executed as a strict linear sequence of blocks: Data Extractor → Specialized AI Agent → Output Formatter.

3.2 Module: Data Extractor (The "Pre-Processor")

- **Target Entities Help:** The Studio provides "AI Suggestions" based on the document category.
- **Common Document Sections (Multi-select):**
  - **A. Generic/Functional Sections:** Introduction/Summary, Analysis of Pros, Analysis of Cons, Conclusions.
  - **B. Domain-Specific Sections:** (Contextual) Financial Statements, Legal Clauses, Technical Specs, etc.

3.3 Module: Specialized AI Agent (The "Reasoning Core")

- **Persona Selection:** Legal Counsel, Financial Auditor, Cybersecurity Architect, Strategic Consultant, Technical Architect.
- **Framework Wizard:** Contextual suggestions (STRIDE for Risk, SWOT for Strategy, TOGAF for Architecture) based on the template type defined at start.

3.4 Module: Variable Management (No-Code Input)

- **Variable Panel:** UI-driven management of dynamic inputs.
- **UI Controls:** Name, Label, Type (Text, Date, Number, List), and Default Value.

3.5 Module: Output Formatter (Target Format)

- **Textual Reports:** Markdown (.md).
- **Diagrams/Schemas:** Mermaid (SVG/PNG).
- **Tables/Data:** CSV or Markdown.

4. Admin Interface: Framework Governance (System Admin)

The Admin Interface allows administrators to manage global settings: Analysis Taxonomy, Knowledge Base Sections, Persona Management, Framework Governance (including AI Specifications), Terminology/Thesauruses, and Logic/UI Metadata.

5. Technical Architecture & Data Objects

5.1 Evidence Objects (JSON Structure)

Every finding must be returned with an array of evidence_objects to ensure absolute traceability (source_id, snippet, page_number, coordinates).

5.2 Map-Reduce Logic

Automatically invoked for large or multiple document sets. Processing is split into a **Map Phase** (individual asset analysis) and a **Reduce Phase** (global synthesis).

6. Critical Development Guardrails

1. **Strict Traceability:** Grounding metadata required for every paragraph.
2. **Expertise Scaling:** Dynamic reasoning depth (Chain of Thought vs. Fast Response).
3. **Hallucination Prevention:** Strict Temperature control (0.0-0.1).
4. **Thesaurus Guard:** Post-processing terminology validation step.

7. JSON Template Schema (The "Workflow" Definition)

This JSON structure is the output of the Template Studio and the input for the Analysis Engine.

7.1 Schema Definition

```
{
 "template_metadata": {
  "id": "uuid",
  "name": "Risk Analysis Audit",
  "category_id": "risk_mgmt",
  "description": "Exhaustive risk audit based on STRIDE framework.",
  "version": "1.0.0"
 },
 "execution_constraints": {
  "min_docs": 1,
  "max_docs": 5,
  "input_mode": "single"
 },
 "user_inputs": [
  {
   "id": "var_target_system",
   "label": "Target System Name",
   "type": "string",
   "default": "Internal Network"
  }
 ],
 "pipeline": [
  {
```

```json
      "step": 1,
      "type": "data_extractor",
      "params": {
        "generic_sections": ["Background", "Analysis of Cons"],
        "domain_sections": ["Technical Specifications", "Risk Assessments"],
        "target_entities": ["Vulnerabilities", "Data Flows", "Assets"],
        "extraction_objectives": "Identify all assets and potential entry points mentioned."
      }
    },
    {
      "step": 2,
      "type": "specialized_ai_agent",
      "params": {
        "persona_id": "cyber_arch",
        "framework_id": "stride",
        "expertise_level_default": "expert",
        "thesaurus_id": "nist_csf",
        "logic_depth": "chain_of_thought",
        "custom_instructions": "Focus specifically on {{var_target_system}} security posture."
      }
    },
    {
      "step": 3,
      "type": "output_formatter",
      "params": {
        "primary_format": "markdown",
        "visual_format": "mermaid",
        "data_format": "csv",
        "include_evidence_table": true
      }
    }
  ],
  "guardrails": {
    "temperature": 0.1,
    "hallucination_check": "strict",
    "enforce_citations": true
  }
}
```

7.2 Exhaustive Risk Analysis Example (Contextual UI)

When a user builds a **Risk Analysis** template:

1. **Framework Wizard**: Suggests *STRIDE*, *DREAD*, or *NIST*. If user selects *STRIDE*, the Admin-defined "AI Specification" for STRIDE is injected into Step 2.
2. **Data Extractor**: Suggests "Risk Assessments" and "Analysis of Cons" as sections.
3. **Agent Persona**: Automatically defaults to "Cybersecurity Architect."
4. **Output**: Automatically configures a "Risk Matrix" (SVG) and an "Evidence List" (.md) as mandatory outputs.

End of Specifications